Science and economics · Worked mark scheme

Physics D.3 · Chemistry S2.3 · Economics 3.6

1. Measuring the curvature of an ion beam

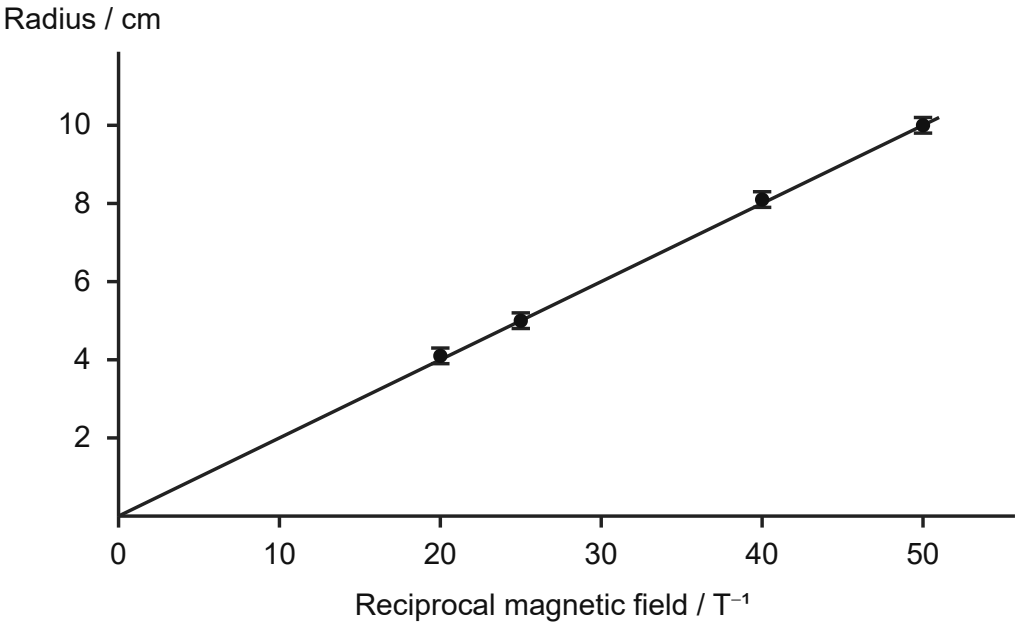
Physics SL / HL · D.3 · Paper 1B-style data analysis · 11 marks

(a) 3 marks

M1 Equate the magnitudes $qvB = mv^2/r$ and rearrange to $q/m = v/(Br)$.

M1 Convert the radius to 0.081 m and the field to 0.0250 T.

A1 1.98×10^8 C kg^{-1}.



The line represents a constant product of 200 mT cm. The error bars show ±0.2 cm.

(b) 3 marks

M1 Compare Br at both extremes using consistent units, for example mT cm.

A1 The intervals are 196–204 and 195–215 mT cm. They overlap.

R1 The measurements are consistent with a constant product within uncertainty. Two readings do not establish a unique law.

(c) 3 marks

A1 Into the page, by the right-hand rule for positive charge.

R1 The magnetic force is perpendicular to the instantaneous velocity.

R1 It does no work, so the kinetic energy and speed stay constant.

(d) 2 marks

A1 For example, increase the calibrated image magnification while keeping the arc in view.

R1 The same physical arc spans more pixels, reducing the fractional effect of locating its edge. Credit a circle fit using many calibrated positions with an appropriate explanation.

2. Finding a camera offset by reversing a field

Physics SL / HL · D.3 · Paper 1B-style data analysis · 12 marks

(a) 3 marks

M1 The offset has the same sign in both readings while the physical displacement changes sign.

A1 Offset is the mean, +0.50 mm.

A1 Field displacement is half the difference, 4.80 mm.

(b) 4 marks

M1 Transit time is $0.0800/(2.00\times10^{5}) = 4.00\times10^{-7}$ s.

M1 Use $y = at^{2}/2$ to obtain $a = 6.00\times10^{10}$ m s^{-2}.

M1 Use $E = 40/0.0200 = 2000$ V m^{-1} and $q/m = a/E$.

A1 3.00×10^{7} C kg^{-1}.

(c) 2 marks

A1 The half-difference at 20 V is 2.40 mm.

R1 At fixed speed and geometry the deflection should be proportional to voltage. Doubling 20 V gives twice the displacement, as observed.

(d) 3 marks

M1 For a half-difference, add the two absolute limits and divide by two.

A1 Maximum uncertainty is 0.10 mm.

R1 Repeats may reduce random scatter but leave a common error in plate separation and therefore in the inferred field.

3. A crossed-field selector with a residual field

Physics SL / HL · D.3 · Paper 1B-style data analysis · 9 marks

(a) 3 marks

M1 The gradient is $(5400 - 2400)/(0.010 - 0.004)$.

A1 5.0×10^5 m s^{-1}.

R1 Zero deflection requires $qE = qvB_{total}$, so the gradient is the selected speed.

(b) 3 marks

M1 The electric-field intercept is $2400 - (5.0 \times 10^5)(0.004) = 400$ V m^{-1}.

M1 Divide the intercept by the selected speed.

A1 8.0×10^{-4} T or 0.80 mT, in the same direction as the applied field.

(c) 3 marks

R1 A constant electric-field zero error also shifts the intercept, so this dataset alone cannot distinguish the explanations.

R1 Measure the actual magnetic field with a separately zeroed probe when the coil current is zero (or independently calibrate the electric-field reading).

R1 A field of about 0.80 mT at zero current supports the magnetic explanation; a zero field directs attention to the electric-field offset.

4. Melting points within an alkali-metal group

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 8 marks

(a) 2 marks

M1 Use $(453.7 - 336.5)/453.7$ multiplied by 100.

A1 25.8%.

(b) 3 marks

A1 The model contains positive ion cores attracted to delocalized electrons.

R1 At the same ion charge and with one outer electron per atom, increasing ion size increases the characteristic separation and weakens attraction.

R1 Less thermal energy is needed to disrupt the ordered lattice, consistent with the lower melting point.

(c) 3 marks

A1 Melting disrupts the fixed, ordered positions of the ion cores.

R1 Mobile delocalized electrons remain in the liquid.

R1 A molten metal can therefore conduct, even though its conductivity may differ from that of the solid.

5. Which observations identify a metallic solid?

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 10 marks

(a) 2 marks

M1 Divide 36.0 by 20.0.

A1 The ratio is 1.80. The liquid current is well above the detection limit.

(b) 3 marks

A1 Mobile delocalized electrons carry charge through the solid.

A1 Layers of positive ion cores can move relative to one another.

R1 Attraction to the delocalized electrons persists as the layers slide, allowing deformation without shattering.

(c) 3 marks

R1 A high melting point alone can also reflect strong ionic attraction.

R1 Q conducts appreciably only after melting, consistent with ions that are fixed in the solid and mobile in the liquid.

A1 A giant ionic lattice is therefore more plausible than a metallic lattice.

(d) 2 marks

A1 A current below 0.01 mA could be non-zero.

R1 A more sensitive measurement would give a lower upper bound, subject to leakage and noise.

6. Resolving two close melting temperatures

Chemistry SL / HL · S2.3 · Paper 1B-style data analysis · 10 marks

(a) 3 marks

A1 Magnesium mean 650 °C, aluminium mean 660 °C.

M1 Half the difference between maximum and minimum.

A1 Both half-ranges are 1.5 °C.

(b) 3 marks

A1 The mean difference is 10 °C and the two observed ranges do not overlap.

R1 Adding the same offset to both series cancels in their difference.

R1 Small scatter indicates repeatability, while a common offset may still affect either absolute temperature.

(c) 4 marks

A1 Aluminium has a higher positive core charge and more delocalized electrons per atom in the model.

R1 Its smaller core and greater charge density support stronger electrostatic attraction, consistent with a higher melting point.

R1 Both charge and size vary in this comparison, so the third electron cannot be isolated as the cause.

R1 The simple model gives a qualitative trend and cannot predict a 10 °C difference from these data alone.

7. Can household transfers close a recessionary gap?

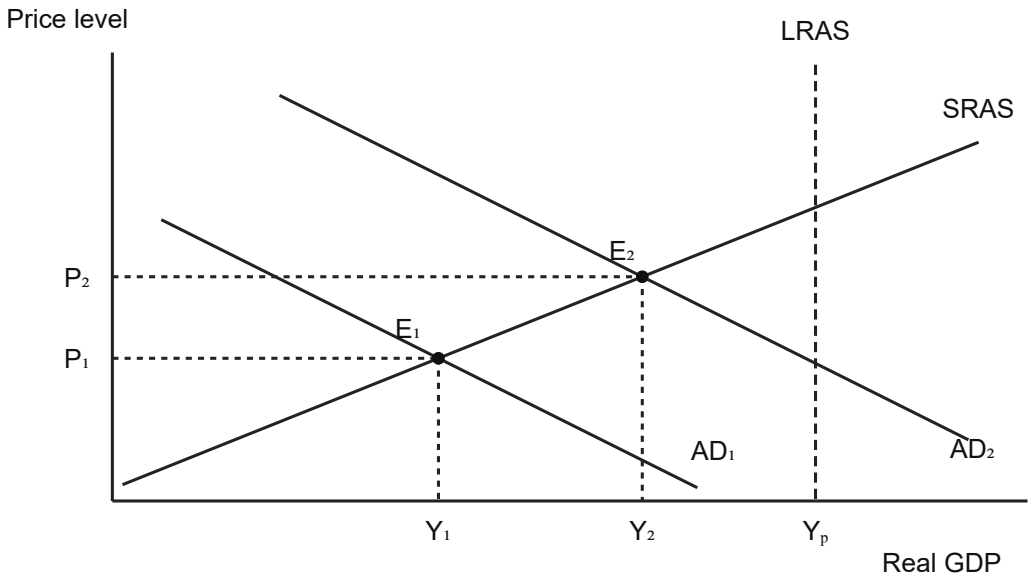
Economics SL / HL · 3.6 · Paper 1 essay option · 25 marks

(a) 10 marks

A recessionary gap exists when actual real output lies below potential output. A transfer raises recipients' disposable income. In the national accounts the government is not purchasing output when it sends the transfer.

If households spend some of the transfer on domestically produced goods and services, consumption rises and aggregate demand shifts right. Firms can employ spare labour and capital. Additional incomes may generate further expenditure, while saving, taxes and imports reduce subsequent domestic rounds.

Draw price level vertically and real GDP horizontally. With upward-sloping SRAS, mark the initial equilibrium below potential GDP. Shift AD to the right and mark the higher equilibrium output and price level. Show potential output and the smaller remaining gap. Credit another internally consistent AD/AS model.



One possible AD/AS diagram for part (a). Potential output is Y_p. Both equilibria lie below it.

Marks	Best-fit descriptor
0	No relevant economic reasoning.
1–2	Identifies a relevant term or consequence with little development.
3–4	Shows partial understanding of transfers or aggregate demand. The chain or diagram is incomplete.
5–6	Develops the link to spending and output with some correct terminology or graphical support.
7–8	Explains the mechanism accurately, using an appropriate diagram and mostly coherent analysis.
9–10	Fully explains how transfers can raise equilibrium output, integrating precise terms, a correctly labelled diagram and the role of spare capacity.

(b) 15 marks

One defensible judgement is that prompt, targeted fiscal support can protect demand during a recession, but the increase in current output depends on what recipients can buy, how much they spend domestically and whether firms can expand supply.

For example, the US CARES Act in March 2020 authorized first-round payments of up to USD 1,200 for eligible individuals. This raised disposable income directly. A household facing a cash shortage could sustain food purchases, limiting lost sales and jobs. Others could save the transfer or repay debt, weakening the immediate increase in consumption.

The initial US BEA release for April 2020 reported disposable personal income rising 12.9% while personal consumption expenditure fell 13.6% from March. Closures constrained purchases of services. These aggregate changes help explain why more income need not produce an immediate spending rise. They cannot identify how much lower spending would have been without the policy.

Direct public procurement adds to government spending immediately when production occurs, while infrastructure projects may face planning lags. Near capacity, extra demand can raise prices more than output. Where spare capacity and domestic suppliers are available, purchases or well-targeted transfers can produce a larger output response. A supply-constrained recession calls for a different timing or mix.

Use the country, timing and policy mechanism of any accurate real-world example. More than one example is welcome. Weigh speed against the domestic spending response, available capacity and the time horizon. Apply the best-fit band to the whole answer.

Marks	Best-fit descriptor
0	No relevant economic reasoning.
1–3	Offers a largely unsupported assertion.
4–6	Describes fiscal measures with limited theory, application or evaluation.
7–9	Develops a relevant economic argument, with a real-world example or evaluation used unevenly.
10–12	Applies accurate theory and a relevant example. Explores competing effects and reaches a reasoned judgement.
13–15	Integrates accurate theory and a developed real-world example throughout. Weighs specific conditions and evidence to reach a convincing judgement about effectiveness. Credit other well-supported conclusions.

Sources

The experimental readings are simulated. The lithium, sodium and potassium melting points are rounded reference values. The economic example uses the estimates available in the May 2020 release.

- IB Physics specimen papers
- IB Chemistry Paper 1B, May 2025
- IB Economics examiner guidance
- Royal Society of Chemistry periodic table
- IRS first-round economic impact payments
- BEA April 2020 personal income and spending release